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MINDPULSE – STUDENT STRESS DETECTION SYSTEM

Insha Ansari[1], Abdullah Ansari[2], Arshi Khan[3], Kashaf Shaikh[4], Farhanaaz Sayyed[5]

[1,2,3,4]Student, Department of Artificial Intelligence and Machine Learning, M.H Saboo Siddik Polytechnic, Mumbai, India.

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ABSTRACT

This study proposes a Student Stress Detection System which has been developed to help students manage the challenges associated with their studies as well as their personal life. The system uses machine learning tools and techniques to monitor in real time stressors that include workload, social engagement, and behaviour patterns. The focus is on preventing stress from worsening, so that educational institutions can provide sufficient help and resources at appropriate times. This proactive method can boost student's mental health and educational attainment. Additionally, the system has features to help avoid burnout and improve the quality of learning. Using this method raises ethical questions like privacy and consent, which are some of the most important matters because student's information has to be treated with caution. Preliminary evidence suggests that this system can enhance student's achievement and mental health.

Keywords: Stress detection, student well-being, machine learning, academic support, mental health, early intervention, data privacy.

1. INTRODUCTION

With social, personal, and academic lives facing greater changes and challenges, stress-related problems among students are on the rise. Student Stress Detection System seeks to address this problem by integrating different types of data including physiological, behavioral, and contextual data to detect early warning signs of stress. It takes it a step ahead by applying machine learning so that users can be provided with real-time insights as well as personalized stress management techniques. This makes students stress-resistant. This tool is designed for the purpose of offering educational institutions and students an ever-evolving solution towards academic as well as mental well-being. Not only does this framework offer ease of integration with other tools of education, but it also offers a convenient interface while ensuring good privacy and security for personal user information.

PROBLEM STATEMENT

Student stress is today a serious issue, affecting their well-being, health, and academic performance. Self-report questionnaires and standard psychological tests are the conventional approaches to identify stress and are not adequate. They rely on student's self-knowledge and the level of willingness to disclose their stress, which will provide inaccurate reports and delayed responses. A more efficient method of stress detection and support for students is required. An integrated Student Stress Detection System has the capability to offer real-time, objective, and automated detection of stress. The system can use physiological measures and behavioural data to offer timely and customized support to students. By implementing such a system, schools can improve the mental health of students in advance, lower stressrelated learning issues, and create a healthier learning environment.

2. IMPELEMNTATION PLAN

Data Privacy and Security

1. Securely preserve all student information while adhering to relevant privacy regulations, such as the GDPR.
2. Student information will only be accessible to users with appropriate permissions and encryption.
3. Facial images of the student's responses, as well as their answers, should be stored under strict confidentiality and integrity.

Machine Learning for System Improvement

1. Employ machine learning models to enhance the accuracy of stress level detection continuously.
2. Use feedback loops so that the system learns over time for better predictive analysis regarding stress levels.

User Interface

1. Create a student-friendly design for the user interface for easy comprehension and interaction.
2. Make the application available on various platforms such as the web, smartphone and computer.
3. Provide seamless navigation for the learners to interact with the questions, post answers and receive feedback.

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Counselor and Instructor Alerts

1. Set up an alert that will inform a teacher or counselor if the student has reached a "High Stress" level threshold.
2. Make sure actionable alerts are given while providing adequate context regarding the stress levels and what possible interventions may be needed. **Cross-Platform Availability**

1. To ensure that students can access the system with ease, it will be available on several devices and platforms.
2. Make sure there's support for different operating systems.

3. SYSTEM DESIGN

The designed framework of the Stress Detection System for Students is aimed towards monitoring student stress levels using real time data collection, processing, analysis, and feedback provision. The input layer employs the use of a webcam to capture student's facial expressions and a questionnaire that captures student responses in real time. This information is pre-processed by verifying facial expression and response accuracy for detail drill down. The information is then relayed into a trained machine learning model that classifies stress levels as high, moderate, or low.

The system checks the captured responses along with the facial features for demonstration of any indications of stress in the analysis layer. Based on this classification, personalized feedback is then given for moderate or high stress which could include relaxation exercises. In cases highest stress is observed, alerts are set off for counsellors or teachers to take action.

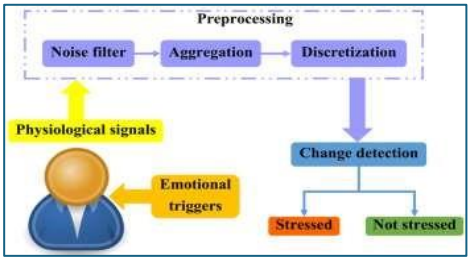


Fig.1: Block Diagram for Student Stress Detection System

Persistent tracking of students is attained by systematic data storage within secure servers and periodic report generation for student wellbeing monitoring over a duration of time. The provided interface is user friendly enabling students to respond to the questions and receive answers with ease and teachers can easily access reports to assist students with further detailed monitoring.

Such a design enables automated and efficient stress detection, allows the educational setting to be addressed in a proactive manner for mental health management.

4. FUTURE SCOPE

With progress in machine learning, wearable technology, and mobile technology, the future scope of the Student Stress Detection System is inflective and deep rooted. Educational institutions can strive to resolve a critical issue of the hour which is prioritizing student's mental health through easing stress signs using the system. Advanced models of machine learning will create additional possibilities for predicting scope enabling more precision to pinpoint stress signals. Dedicated worrying interventions and responses will be provided as the system determines how students in real time respond which is one area that is bound to receive attention. Considering sleep, emotional state, and environmental quality as additional psychological factors will allow for a better understanding of mental health of students. Furthermore, students can benefit from the convenience of mobile apps to allow for monitoring on the go. Educational institutions can then incorporate the ease and assistance on these technologies into comprehensive wellness platforms in order to adopt a more preventive approach to look after student's health, both physical and mental, and academic performance.

5. CONCLUSION

In general, the Student Stress Detection System offers a new paradigm to help students maximize their individual well being and academic success. The system integrates a range of technologies such as machine learning, real time data processing, and behaviour analysis to enable steps of stress detection and intervention to be offered in real time. With more and more schools focusing on mental health, educators and school counsellors can just identify students in need and give them proper feedback that keeps stress levels in check. The system can also include technological advancements with mobile phones and wearables, and thus it is a crucial element towards enhancing the learning environment. The Student Stress Detection System treats mental health and accomplishments at the same time, thus it is a significant aspect in offering sufficient educational circumstances.

6. REFERENCES

[1] J. S. et al., "College Enquiry Chatbot," 2023 2nd International Conference on Vision Towards Emerging Trends in Communication and Networking Technologies (ViTECoN), pp. 1-4, 2023, doi: 10.1109/ViTECoN58111.2023.10156978.

[2] H. K. K. et al., "Smart College Chatbot using ML and Python," 2020 International Conference on System, Computation, Automation and Networking (ICSCAN), pp. 1-5, 2020, doi: 10.1109/ICSCAN49426.2020.9262426.

[3] R. Parkar et al., "AI And Web-Based Interactive College Enquiry Chatbot," 2021 13th International Conference on Electronics, Computers and Artificial Intelligence (ECAI), pp. 1-5, 2021, doi: 10.1109/ECAI52376.2021.9515065.

[4] G. S. Sai Vikas et al., "Information Chatbot for College Management System Using Multinomial Naive Bayes," 2021 2nd International Conference on Smart Electronics and Communication (ICOSEC), pp. 1149-1153, 2021, doi: 10.1109/ICOSEC51865.2021.9591757.

[5] A. Mondal et al., "Chatbot: An automated conversation system for the educational domain," 2018 International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP), pp. 1-5, 2018, doi: 10.1109/iSAI-NLP.2018.8692927.

[6] S. Kesarwani et al., "Student Chatbot System: A Review on Educational Chatbot," 2023 7th International Conference on Trends in Electronics and Informatics (ICOEI), pp. 1578-1583, 2023, doi: 10.1109/ICOEI56765.2023.10125876.

[7] A. Bhharathee et al., "AI-Powered Student Assistance Chatbot," 2023 International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT), pp. 487-492, 2023, doi: 10.1109/IDCIoT56793.2023.10053439.

[8] G. Mallikarjuna Rao et al., "Smart-Bot Assistant for College Information System," 2022 Second International Conference on Artificial Intelligence and Smart Energy (ICAIS), pp. 693-697, 2022, doi: 10.1109/ICAIS53314.2022.9742871.

[9] X. Gonsalves et al., "Designing an Interactive Chatbot for Educational Assistance using Rasa Framework," 2023 International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT), pp. 68-74, 2023, doi: 10.1109/IDCIoT56793.2023.10053457.

[10] B. R. Ranoliya et al., "Chatbot for university related FAQs," 2017 International Conference on Advances in Computing, Communications and Informatics (ICACCI), pp. 1525-1530, 2017, doi: 10.1109/ICACCI.2017.8126057.

[11] A. K. Nikhath et al., "An Intelligent College Enquiry Bot using NLP and Deep Learning based techniques," 2022 International Conference for Advancement in Technology (ICONAT), pp. 1-6, 2022, doi: 10.1109/ICONAT53423.2022.9725865.

[12] R. R. Anumala et al., "Execution of College Enquiry Chatbot using IBM virtual Assistant," 2022 International Conference on Computing, Communication and Power Technology (IC3P), pp. 242-245, 2022, doi: 10.1109/IC3P52835.2022.00058.

[13] D. P. P. Villanueva et al., "A Chatbot as a Support System for Educational Institutions," 2021 62nd International Scientific Conference on Information Technology and Management Science of Riga Technical University (ITMS), pp. 1-6, 2021, doi: 10.1109/ITMS52826.2021.9615271.

[14] W. Mahanan et al., "College Agent: The Machine Learning Chatbot for College Tasks," 2022 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECT).

[15] S. Meshram et al., "College Enquiry Chatbot using Rasa Framework," 2021 Asian Conference on Innovation in Technology (ASIANCON), pp. 1-8, 2021, doi: 10.1109/ASIANCON51346.2021.9544650.

www.ijprems.com (Int Peer Reviewed Journal) Factor : editor@ijprems.com Vol. 05, Issue 03, March 2025, pp : 673-676 7.001

[16] M. -T. Nguyen et al., "Building a Chatbot for Supporting the Admission of Universities," 2021 13th International Conference on Knowledge and Systems Engineering (KSE), pp. 1-6, 2021, doi: 10.1109/KSE53942.2021.9648677.

[17] Elashmawi et al., "An Interactive Chatbot for College Enquiry," 2023.

[18] S. Kumari et al., "Enhancing College Chat Bot Assistant with the Help of Richer Human Computer Interaction and Speech Recognition," 2020 International Conference on Electronics and Sustainable Communication Systems (ICESC), pp. 427-433, 2020, doi: 10.1109/ICESC48915.2020.9155951.

[19] N. P. Patel et al., "AI and Web-Based Human-Like Interactive University Chatbot (UNIBOT)," 2019 3rd International conference on Electronics, Communication and Aerospace Technology (ICECA), pp. 148-150, 2019, doi: 10.1109/ICECA.2019.8822176.

[20] V. Singh et al., "ChatBot using Python Flask," 2023 7th International Conference on Intelligent Computing and Control Systems (ICICCS), pp. 1182-1185, 2023, doi: 10.1109/ICICCS56967.2023.10142484.

[21] L. Tunstall et al., "Natural Language Processing with Transformers: Building Language Applications with Hugging Face," O'Reilly Media, 2022.

[22] R. Arumugam et al., "Hands-On Natural Language Processing with Python," Packt Publishing, 2018.

[23] S. Kublik et al., "GPT-3: Building Innovative NLP Products using OpenAI's Powerful Language Model," Independently published, 2021.

[24] OpenAI Developer Channel, "OpenAI API Quickstart - Build Your First GPT-3 App," YouTube.

[25] Tech With Tim, "Creating AI Chatbots Using OpenAI and GPT-3," YouTube.

[26] OpenAI Blog, "OpenAI API GPT-3 Use Cases," OpenAI, 2023.

[27] Towards Data Science, "How to Build a Chatbot with GPT-3," Towards Data Science, 2023.

[28] Hugging Face Blog, "Building NLP Applications with Transformers," Hugging Face, 2023.

[29] Analytics Vidhya, "Develop Your First AI Chatbot with Python and NLP," Analytics Vidhya, 2023.

[30] LangChain Blog, "Building a GPT-powered Chatbot with LangChain," LangChain, 2023.

[31] Shivam et al., "Chatbot for College Website," 2018.

[32] S. Rohan et al., "Design and Development of College Chatbot," 2020.